
AGENTIC AI CAPSTONE PROJECT

Agro-Mind

Autonomous AI Agent for Agricultural Support

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0.1 Abstract

Agro-Mind is an autonomous AI-based agricultural customer support agent developed as a capstone project for the Saudi Digital Academy in collaboration with WeCloud-Data. The system aims to provide intelligent and scalable agricultural assistance by integrating Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), computer vision-based crop disease diagnosis, and agentic workflow orchestration using LangGraph.

The system processes customer requests through an intelligent workflow that includes user intent classification, agricultural knowledge retrieval, crop image analysis, product recommendation, customer context handling, and safety-based human escalation. The agent uses RAG to retrieve relevant information from agricultural documents, reducing unsupported responses and improving answer reliability. The vision-based diagnosis module analyzes uploaded crop images to identify potential diseases and provide recommended actions. The system was evaluated using 30+ real-world agricultural scenarios covering text queries, image diagnosis, product inquiries, and safety-sensitive cases. After optimization of the retrieval pipeline, agent decision flow, and product recommendation module, Agro-Mind achieved a 100% success rate across all tested workflows, including intent detection, diagnosis, recommendations, safety validation, and escalation handling.

The project demonstrates the effectiveness of agentic AI in automating agricultural customer support while maintaining accuracy, reliability, and controlled decision-making. The final implementation shows how LLM reasoning, RAG-based knowledge grounding, and intelligent tool execution can be combined to deliver a practical AI assistant for agricultural users.

Keywords: Agentic AI, LangGraph, RAG, Crop Disease Diagnosis, Agricultural Support, LLM, Customer Support Automation

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Introduction

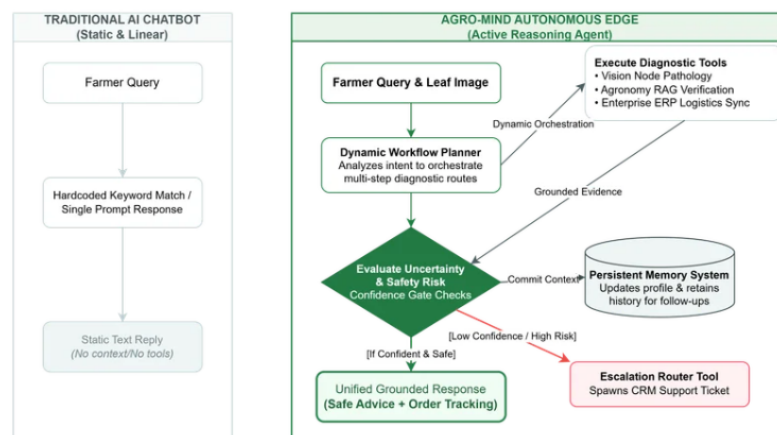
1.1 Project Background

Agriculture plays an important role in food production and economic development. However, farmers and agricultural customers often require fast and accurate support for crop diseases, pests, pesticide usage, and product recommendations. Traditional customer support systems depend mainly on human agents, which makes handling a large number of agricultural inquiries difficult and time-consuming.

Agro-Mind is an autonomous AI-based customer support agent designed to assist agricultural users by combining Large Language Models (LLMs), image-based crop diagnosis, Retrieval-Augmented Generation (RAG), and intelligent tool usage. The system helps users analyze crop problems, retrieve agricultural knowledge, recommend suitable products, answer questions, and escalate sensitive cases to human experts when required.

The system supports the complete customer lifecycle, starting from image upload and disease diagnosis, followed by recommendations, purchase support, logistics inquiries, and follow-up interactions.

Agro-Mind: Active Reasoning vs. Static Chatbots



1.2 Problem Statement

According to the Food and Agriculture Organization (FAO), up to 40% of global crop production is lost every year due to plant pests and diseases, causing economic losses exceeding \$220 billion annually. These losses highlight the importance of providing farmers and agricultural customers with fast, accurate, and reliable support to identify crop problems early and apply appropriate solutions.

Agricultural customer support presents unique challenges because user inquiries are often visual in nature, domain-specific, safety-sensitive, and require expert-level knowledge. Farmers may upload images of unhealthy crops, ask about specific diseases, request pesticide dosage information, or require guidance on product selection.

Existing support workflows face several critical limitations:

- Slow response times due to high inquiry volume and dependence on human availability.
- Repeated manual diagnosis of similar crop problems, reducing efficiency.
- Inconsistent recommendations across different support agents.
- Lack of customer history tracking, preventing personalized support.
- Risk of incorrect pesticide guidance, which may cause safety hazards or crop damage.
- Inability to scale during peak agricultural seasons.

Therefore, an intelligent system is required to automate agricultural support by classifying user requests, analyzing crop images, retrieving reliable agricultural information, providing context-aware recommendations, and transferring critical or high-risk cases to human experts when necessary.

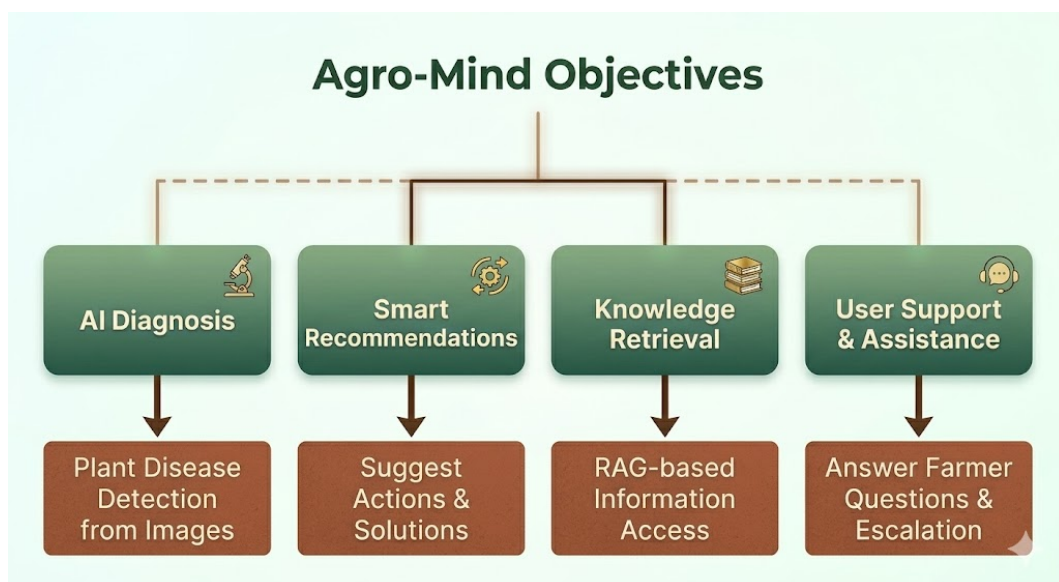
1.3 Project Objectives

The main objective of Agro-Mind is to develop an autonomous AI support agent that provides reliable agricultural assistance and improves customer support efficiency.

The system aims to:

- Allow users to upload crop images and receive possible disease or pest diagnosis.
- Automatically classify user requests using Large Language Models.
- Retrieve agricultural information from a knowledge base using RAG technology.
- Provide product recommendations and safe usage guidance.
- Answer agricultural questions related to crops, diseases, and pesticides.
- Detect unsafe cases and escalate them to human support agents.
- Track customer interactions and maintain user profiles for future support.
- Generate summaries and reports to assist agricultural support teams.

Objectives Blocks Diagram



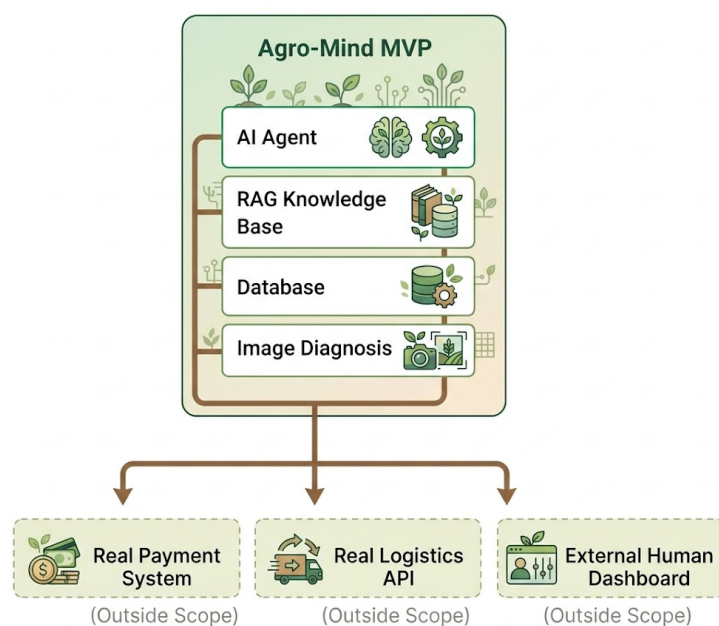
1.4 Project Scope

The scope of Agro-Mind focuses on developing an AI-powered agricultural support system that combines conversational AI, image analysis, knowledge retrieval, and customer management. The system includes a chat interface allowing users to send text messages and upload crop images. The AI agent analyzes user intent, performs diagnosis when needed, retrieves information from the agricultural knowledge base, recommends suitable actions, and updates customer history.

The project includes the following components:

- AI Agent workflow powered by LangGraph
- Crop image diagnosis using computer vision
- RAG-based agricultural knowledge retrieval
- Product recommendation module
- Customer profile management system
- Human escalation system for high-risk cases
- Logging and monitoring infrastructure

Scope Boundary Diagram:



System Design

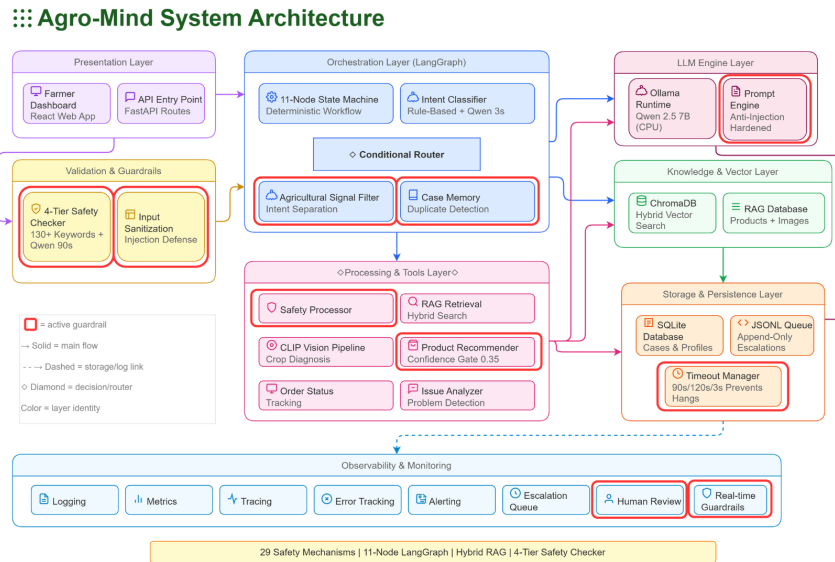
2.1 Overall Architecture

The Agro-Mind system follows an agent-based architecture where users interact with the system through a conversational interface. User requests, including text queries and crop images, are processed by the Agro-Mind AI Agent, which acts as the central orchestration component responsible for understanding user intent and selecting the appropriate workflow.

The agent coordinates different AI modules and services, including image diagnosis, Retrieval-Augmented Generation (RAG), product recommendation, customer memory management, safety verification, and human escalation. This enables the system to provide accurate, context-aware, and reliable agricultural support.

The main architecture components are:

- **Chat Interface:** Provides user interaction and supports text queries and crop image uploads.
- **Agro-Mind AI Agent:** Acts as the decision-making layer that controls the workflow execution and coordinates available tools.
- **Image Diagnosis Module:** Analyzes uploaded crop images to detect possible diseases, pests, or plant-related issues.
- **RAG Knowledge Base:** Retrieves relevant agricultural information from curated documents to enhance response accuracy and reduce hallucination.
- **Database Layer:** Stores customer profiles, conversations, diagnostic results, recommendations, and system history.
- **Human Escalation Module:** Handles high-risk, sensitive, or complex cases by routing them to agricultural experts.



2.1.1 Deployment Architecture

The Agro-Mind system is implemented using a layered deployment architecture that separates user interaction, processing logic, AI orchestration, and data management components.

- **Frontend Layer:** Provides the user interface for interacting with the system and uploading crop images.
- **Backend Service Layer:** Uses FastAPI to manage API communication between the frontend, AI components, and databases.
- **AI Orchestration Layer:** Uses LangGraph to control the agent workflow, tool execution, and decision-making process.
- **Vector Database Layer:** Uses ChromaDB to store embeddings and perform semantic retrieval for the RAG pipeline.
- **Storage Layer:** Uses SQLite for persistent storage of user information, conversations, and system records.
- **LLM Inference Layer:** Uses Ollama to host and serve the language model responsible for generating final responses.

This layered architecture improves modularity, simplifies system maintenance, and supports future scalability and integration with additional agricultural services.

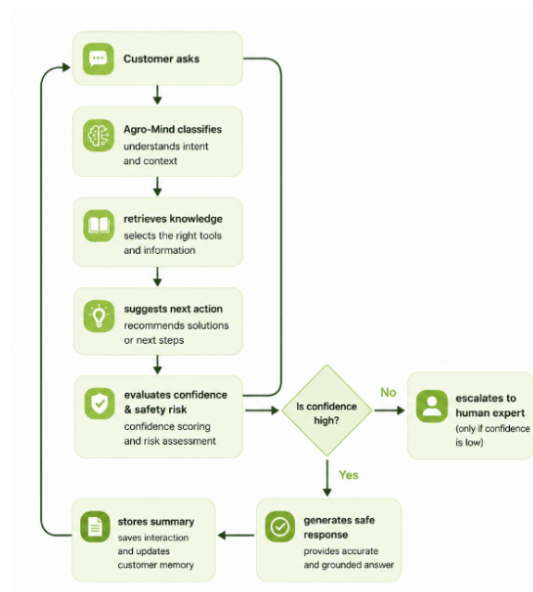
2.2 Agro-Mind Agent Workflow

The Agro-Mind Agent is the core decision-making component of the system. Unlike a simple chatbot, the agent analyzes each user request and selects appropriate tools based on the inquiry type. When a farmer uploads a crop image, the agent identifies the diagnosis intent, analyzes the image, retrieves related agricultural knowledge, and generates recommendations. For safety-sensitive requests, the agent performs a safety check and escalates the case to human support when required.

The main workflow steps are:

1. Receive user message or crop image.
2. Load customer profile from the database.
3. Classify user intent using the LLM.
4. Run the safety checker to detect high-risk inputs.
5. Route to the appropriate workflow path (knowledge retrieval, image diagnosis, order support, or escalation).
6. Generate response using the LLM with retrieved context and safety enforcement.
7. Save the support case and update the customer profile.

Agent Workflow Diagram



2.3 Database Design

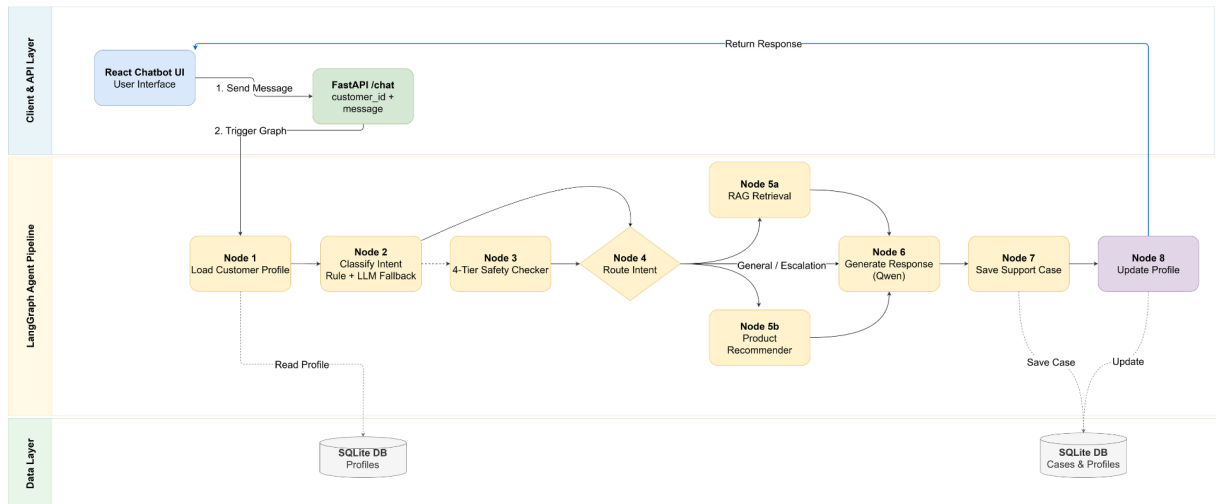
The Agro-Mind system uses a multi-layer database design to store customer information, conversations, diagnosis results, recommendations, and operational data. The database supports customer lifecycle tracking and enables personalized future recommendations.

The system maintains two types of memory:

- Short-term memory — stores current conversation context, active session data, and real-time diagnosis state.
- Long-term memory — stores customer history including previous diagnoses, purchases, and interactions, implemented using a ChromaDB vector database (chromadb/) for semantic retrieval.

Structured operational records are stored in JSONL files, including customer profiles (customers.jsonl) and escalation cases (escalations.jsonl).

Image-based diagnosis is supported by a dedicated vector store (vision_chromadb) which indexes image embeddings using an HNSW index for fast similarity search.



Data Flow Diagram

2.3.1 Database Schema

The Agro-Mind database consists of several interconnected entities that enable the system to manage agricultural support operations effectively. Customer profiles store user and farm-related information, while conversation records maintain interaction history and session summaries. Diagnostic entities capture image analysis results and confidence scores, and recommendation entities track suggested products and actions. Additional entities support order management, logistics tracking, and the escalation of complex or high-risk cases to human agronomists, ensuring reliable and scalable customer support.

The main entities include:

- Customers - Stores user profiles and agricultural information
- Conversations - Stores chat history and session summaries
- Diagnoses - Stores image diagnosis results and confidence levels
- Products - Stores agricultural product catalog information
- Recommendations - Stores suggested products and actions per case
- Orders - Stores logistics and order tracking information
- Escalations - Stores high-risk cases transferred to human agents

These entities are linked through customer and case identifiers to ensure traceability throughout the customer support lifecycle.

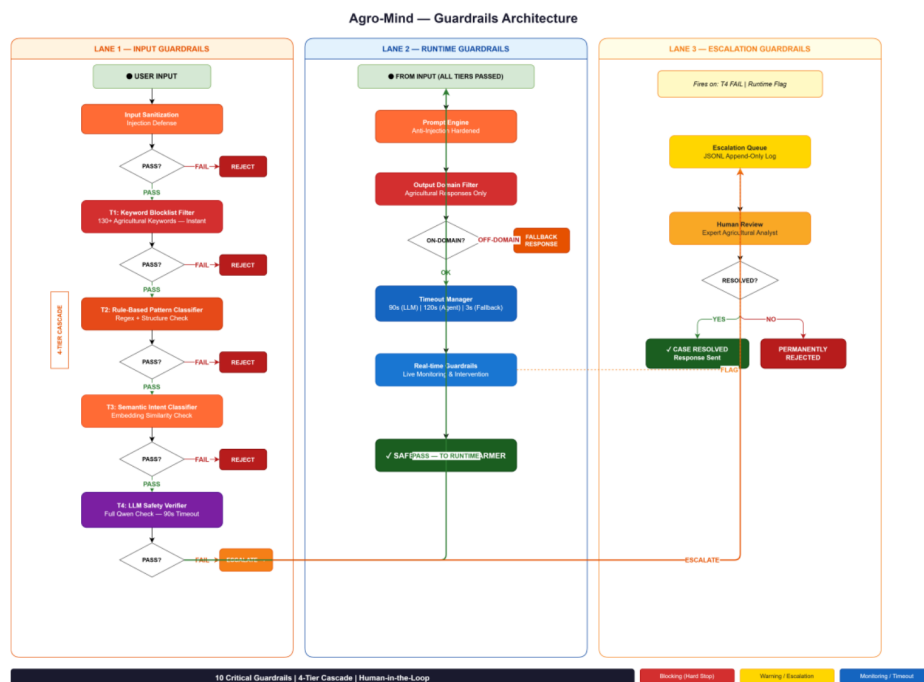
2.4 Human Escalation and Monitoring

Within the Agro-Mind Guardrails Architecture, Human Escalation acts as the final safety layer after all input and runtime guardrails have been evaluated. Requests first pass through multiple validation stages, including input sanitization, toxicity detection, prompt injection protection, semantic risk classification, and runtime safety monitoring. The LLM Safety Verifier then performs a final assessment to determine whether the system can safely generate a response.

If the request is considered safe, the agent continues normal processing and generates a response. However, if the LLM Safety Verifier detects a high-risk, ambiguous, or policy-sensitive case, the request is flagged for escalation. In such cases, the system immediately stops further processing, bypasses retrieval and recommendation generation, and routes the case to the Human Escalation Queue for expert review.

The monitoring component continuously records guardrail decisions, escalation events, runtime interventions, and conversation logs. These records support auditing, performance evaluation, and continuous improvement of the Agro-Mind Guardrails Architecture.

Guardrails Architecture



Implementation

3.1 Technology Stack

The Agro-Mind system was developed using AI and web technologies to provide intelligent agricultural customer support. The backend was built with Python and FastAPI, while LangGraph manages the AI agent workflow and task execution. LLMs are used for intent classification, reasoning, and response generation. A RAG pipeline retrieves agricultural knowledge for disease diagnosis, product recommendations, and safety guidance. The system also includes image analysis and a database for storing customer data, conversations, diagnoses, and escalation records.

Technologies used:

Layer	Technology	Purpose
Backend Framework	Python + FastAPI	API handling and service orchestration
Agent Workflow	LangGraph / LangChain	Agentic decision-making and tool calling
Language Model	Qwen via Ollama	Intent classification and response generation
Knowledge Retrieval	ChromaDB	Vector-based RAG knowledge base
Image Diagnosis	Vision Model (ViT-B/32)	Crop disease detection from images
Database	SQLite	Structured data and customer profiles
Frontend Interface	React	Chat interface and image upload UI
Monitoring	LangSmith	Agent tracing, debugging, and evaluation

3.1.1 LLM Configuration

Agro-Mind uses the Qwen2.5-7B - instruct model through Ollama for local inference.

The model was selected because it provides:

- Strong reasoning capabilities.
- Efficient local deployment.
- Low operational cost.
- Good multilingual support.

The model is configured with controlled generation parameters to ensure consistent and reliable agricultural recommendations.

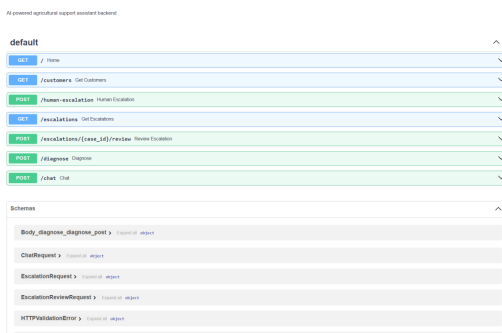
3.2 Frontend and Backend Implementation

The Agro-Mind frontend provides a chat-based interface allowing farmers and agricultural customers to submit text questions and upload crop images. The interface displays diagnosis results, product recommendations, and support responses in real time.

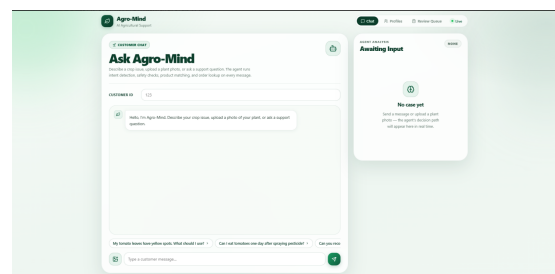
The backend is built using FastAPI and is responsible for processing user requests, managing agent execution, connecting to external tools, and persisting system information. The frontend communicates with the FastAPI backend through REST API endpoints. The backend forwards user queries to the Agro-Mind agent, which determines the required workflow — diagnosis, knowledge retrieval, recommendation, or human escalation.

The implementation supports the following interaction types:

- Text-based conversational queries
- Crop image upload for disease diagnosis
- AI-generated responses and recommendations
- Real-time diagnosis results with confidence scores
- Customer profile creation and update
- Escalation notifications and summaries



Backend



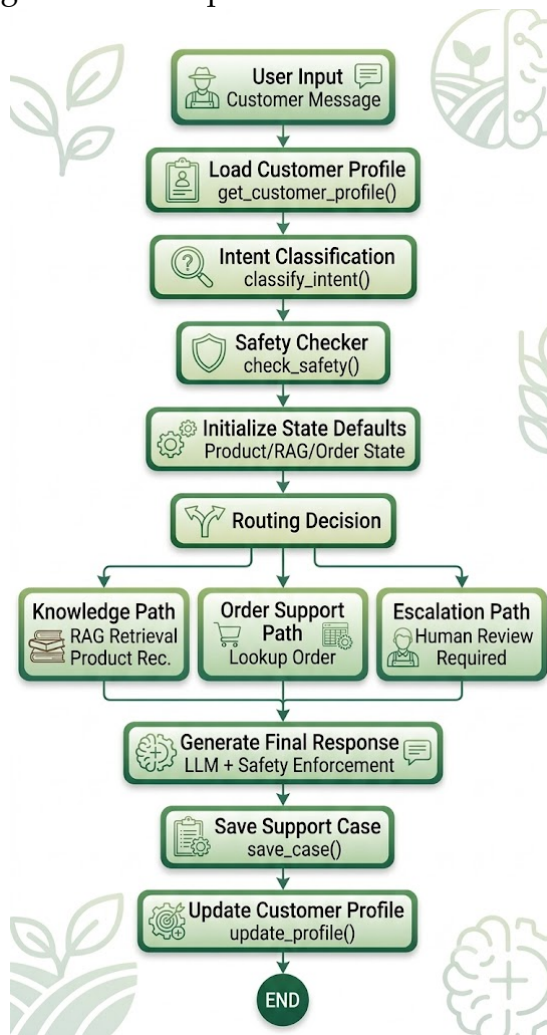
Frontend

3.3 LangGraph Agent Implementation

The core intelligence of Agro-Mind is implemented using LangGraph, which creates an agentic workflow capable of making autonomous decisions. Unlike a fixed chatbot flow, the LangGraph agent analyzes each user request and determines which actions to execute. It can classify the user's intent, call required tools, retrieve information, generate responses, and escalate critical cases.

LangGraph manages the conversation state and ensures each step is executed in the correct order with proper error handling and fallback paths.

LangGraph Workflow Diagram



3.3.1 Tool Specifications

Agro-Mind utilizes a set of modular tools integrated within the LangGraph workflow, where each tool is responsible for performing a specific task within the agricultural support pipeline.

Each tool operates independently and returns structured outputs that are consumed by the workflow orchestrator. These outputs help LangGraph determine the next execution step and maintain an efficient decision-making process.

Table 3.1: Agro-Mind Tool Specifications

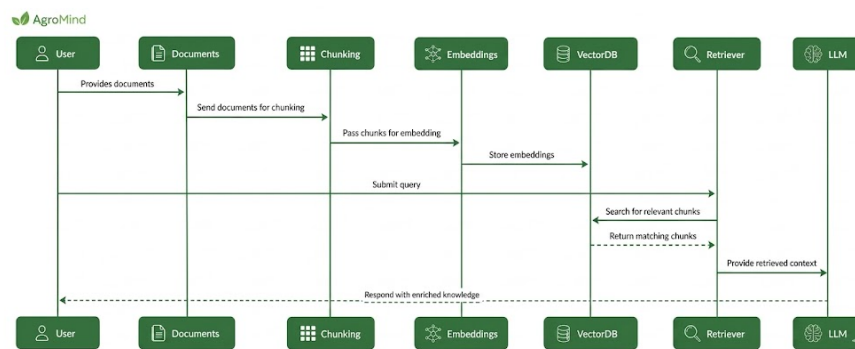
Tool	Purpose
<code>classify_intent()</code>	Detects the category of the user's request.
<code>diagnose_image()</code>	Identifies crop diseases from uploaded images.
<code>retrieve_knowledge()</code>	Retrieves relevant agricultural information using RAG.
<code>recommend_products()</code>	Generates product recommendations.
<code>check_safety()</code>	Evaluates potential safety risks.
<code>update_customer_profile()</code>	Updates customer history and memory.
<code>escalate_to_human()</code>	Routes high-risk cases to human experts.

3.4 RAG and Knowledge Retrieval

The Agro-Mind system uses Retrieval-Augmented Generation (RAG) to provide accurate agricultural responses grounded in trusted knowledge sources. The knowledge base contains crop disease information, pesticide usage instructions, product details, safety guidelines, and frequently asked questions.

When a user asks an agricultural question, the system retrieves the most relevant document chunks from the vector database using semantic similarity search. The retrieved information is then provided to the LLM as context, enabling it to generate grounded responses and significantly reducing hallucination.

RAG Architecture Diagram



3.4.1 RAG Implementation Details

The Retrieval-Augmented Generation (RAG) module uses ChromaDB as the vector database to store agricultural knowledge documents. The retrieval workflow consists of the following steps:

1. Document preprocessing and chunking.
2. Embedding generation for each document chunk.
3. Similarity search against the vector database.
4. Retrieval of the most relevant knowledge chunks.
5. Grounding the LLM response using the retrieved information.

To improve response quality, only the highest-ranked document chunks are provided to the language model. This approach reduces hallucinations and ensures that recommendations are generated based on verified agricultural information.

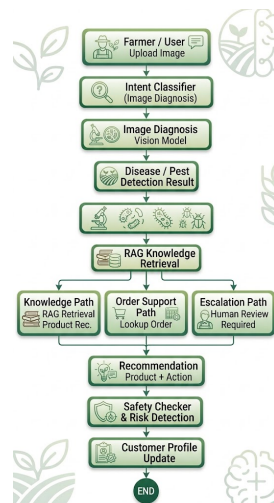
3.5 Image Diagnosis System

The image diagnosis module enables farmers to receive AI-assisted analysis of crop health from uploaded photographs. When a farmer uploads an image, the agent first classifies the user intent to confirm that image-based diagnosis is required.

The image is then processed by the vision model (ViT-B/32) to identify possible crop diseases or pest symptoms and generate a confidence score. The diagnosis pipeline proceeds as follows:

- Farmer uploads a crop image through the chat interface.
- The intent classifier identifies the request as an image diagnosis task.
- The image diagnosis module processes the image and generates a disease/pest detection result with a confidence score.
- The RAG module retrieves related agricultural knowledge based on the diagnosis.
- The system determines the appropriate path (knowledge, order support, or escalation).
- The safety checker verifies that recommendations comply with pesticide safety guidelines.
- A final response is generated with grounded explanations and recommended next steps.
- The diagnosis result and interaction history are stored in the customer profile.

Image Diagnosis Pipeline



3.6 Safety Checker and Customer Profile

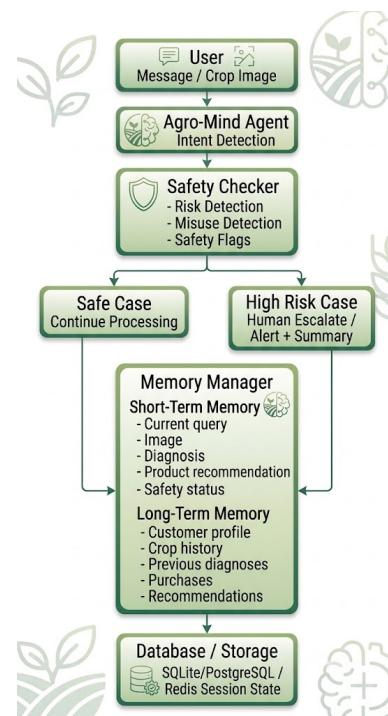
The safety checker module monitors all user requests to identify unsafe pesticide usage, poisoning risks, or situations requiring human expertise. If a high-risk case is detected, the agent halts the normal recommendation flow and creates an escalation request for a human support agent, including a comprehensive case summary.

The customer profile system provides long-term memory by storing customer history, previous diagnoses, purchases, recommendations, and interaction records. This information enables the system to provide personalized responses and context-aware future recommendations.

The customer profile stores:

- Customer identification and contact information
- Types of crops managed by the customer
- Previous diagnosis results and confidence levels
- Product recommendations and purchase history
- Full conversation history and summaries
- Escalation records and resolution status

Safety and Memory Flow



Testing and Results

4.1 Testing Approach

The testing process was designed to verify the functionality, reliability, and safety of the Agro-Mind system. The testing strategy focused on the main AI workflows, including user intent classification, crop image diagnosis, knowledge retrieval, product recommendation, safety checking, and human escalation.

Three testing levels were applied:

- Functional Testing — validating individual system components and tools in isolation.
- Integration Testing — verifying communication between the agent, database, RAG system, vision model, and external tools.
- User Scenario Testing — evaluating end-to-end customer support workflows using realistic agricultural scenarios.

A total of 30 test cases were created to simulate real agricultural customer interactions, including text-based questions, crop image uploads, product inquiries, order tracking requests, and unsafe/high-risk scenarios.

Testing Workflow Diagram



4.2 System Evaluation

The Agro-Mind system was evaluated to verify its ability to operate as an autonomous agricultural support agent. The evaluation focused on intent classification, image diagnosis, knowledge retrieval, recommendation generation, safety control, and customer memory.

The system was tested using realistic agricultural scenarios to verify whether the agent could understand user requests, select the appropriate workflow, process required information, and generate suitable responses.

Intent classification was evaluated using different queries related to diagnosis, products, orders, general questions, and safety cases. The agent successfully identified user intents and routed requests to the correct workflows.

The image diagnosis workflow was tested using crop images with visible symptoms. The system analyzed images, retrieved relevant agricultural knowledge, and provided possible disease information with recommended actions while considering uncertainty.

The RAG component was evaluated using agricultural questions about diseases, pests, and products. Results showed that the system retrieved relevant information and generated reliable responses with reduced hallucination.

Product recommendation was tested based on diagnosis results and customer context. The system generated relevant suggestions while applying safety constraints.

Safety, escalation, and memory management were also evaluated. The system detected high-risk requests, avoided unsafe responses, escalated critical cases, and used stored customer context to provide personalized support.

4.2.1 Error Analysis

Although all evaluation test cases passed successfully, several challenging scenarios were identified during development. These included ambiguous crop images, incomplete user descriptions, and low-confidence diagnoses.

To address these situations, Agro-Mind employs confidence thresholds, clarification requests, and human escalation mechanisms to ensure safe and reliable recommendations.

Capability	Test Description	Outcome
Intent Classification	Varied user queries including diagnosis, product, order, and general questions	Agent correctly routed each request, no misclassification in basic scenarios
Image Diagnosis	Uploaded crop images with visible disease symptoms	System identified possible diseases with confidence scores; recommended follow-up with agronomist
RAG Knowledge Retrieval	Agricultural questions on crop diseases, pests, and pesticide usage	Relevant documents retrieved; grounded responses with reduced hallucination
Product Recommendation	Requests for product suggestions based on diagnosed problems	Relevant products recommended with safety checks applied
Safety & Escalation	High-risk inputs including unsafe pesticide queries	High-risk cases detected and escalated; no unsafe instructions provided
Customer Memory	Multi-turn conversations referencing previous diagnoses	Profile updated correctly; personalized context used in follow-up responses

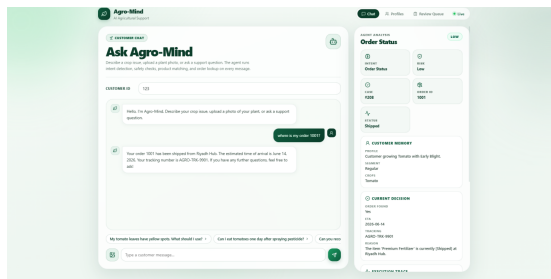
Evaluation Result

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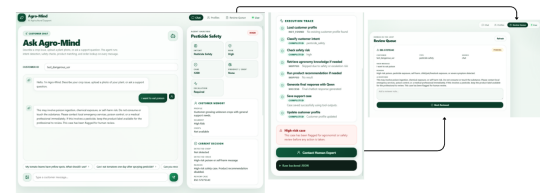
system_benchmark.py
test_case_memory.py
test_conversation.py
test_customer_profile.py
test_escalation_queue.py
test_graph.py
test_human_escalation.py
test_image.py
test_intent_classifier.py
test_llm_agent.py
test_logistics_lookup.py
test_product.py
test_safety_checker.py
test_vision_diagnosis.py

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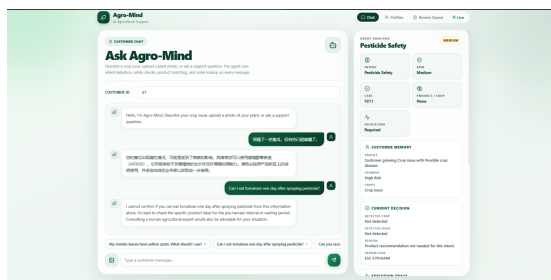
Test Files



Order Case



Human Escalation Case



Normal Case & Chinese , English

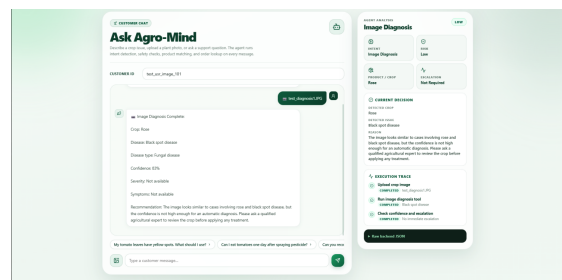


Image Diagnosis

4.3 Results Discussion

The results confirmed that Agro-Mind successfully performs the main customer support workflow by combining LLM reasoning, Retrieval-Augmented Generation (RAG), image analysis, and intelligent tool calling.

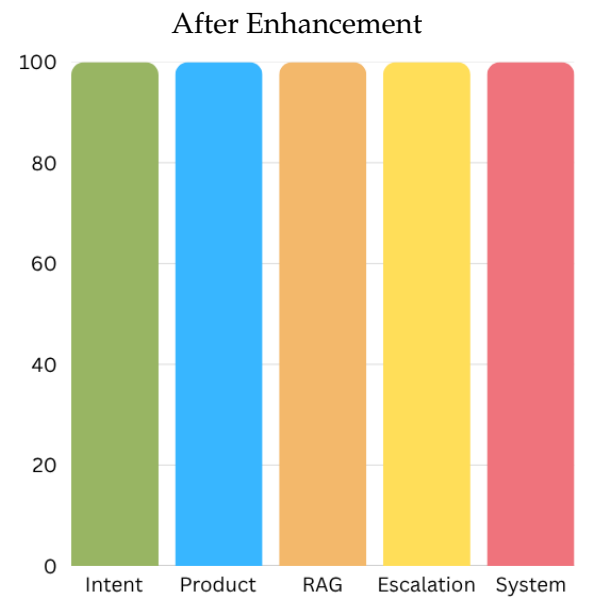
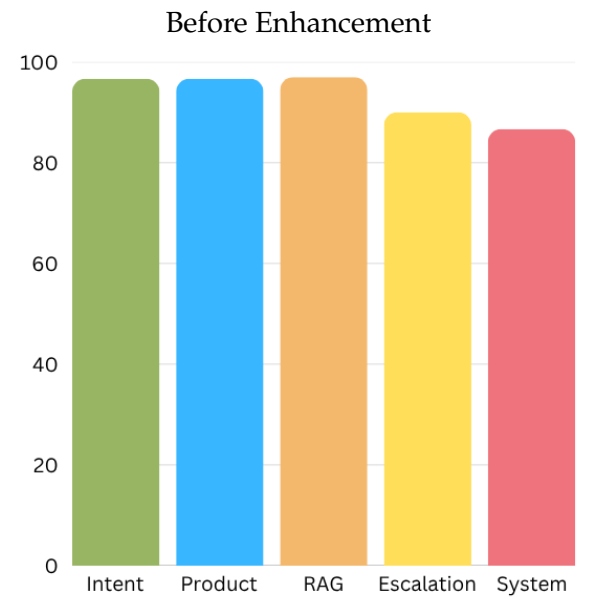
The RAG module improved response reliability by grounding answers in agricultural documents rather than relying solely on the LLM's parametric knowledge. The image diagnosis workflow accurately identified crop diseases, provided confidence scores, and generated appropriate follow-up recommendations. The safety mechanism successfully detected all high-risk test cases and correctly redirected users to human support when necessary.

Following several optimization phases, the system showed measurable improvements across multiple components. Enhancements to the retrieval pipeline increased the relevance of retrieved agricultural knowledge, while refinements to the agent workflow improved decision-making consistency and tool selection. In addition, expanding and restructuring the product catalog eliminated previous recommendation mismatches, increasing product recommendation accuracy from 86.67% to 100%.

The final evaluation achieved a success rate of 100% across all tested scenarios, including intent classification, disease diagnosis, knowledge retrieval, product recommendation, safety validation, and human escalation handling. These results demonstrate that the optimized Agro-Mind system satisfies the functional requirements of the project and provides reliable AI-powered agricultural customer support.

FINAL RESULTS	
Total Tests:	30
Passed Tests:	26/30
Intent Accuracy:	96.67%
Product Accuracy:	96.67%
RAG Accuracy:	96.67%
Escalation Accuracy:	90.00%
System Accuracy:	86.67%
Average Latency:	3.12 sec

FINAL RESULTS	
Total Tests:	30
Passed Tests:	30/30
Intent Accuracy:	100.00%
Product Accuracy:	100.00%
RAG Accuracy:	100.00%
Escalation Accuracy:	100.00%
System Accuracy:	100.00%
Average Latency:	2.95 sec



Before

After

Metric	Value	Notes
Total Test Cases	30	Covering all major workflow scenarios
Passed Tests	26 / 30	4 failures in edge-case product matching
Intent Classification Accuracy	96.67%	Strong performance across all intent categories
RAG Retrieval Accuracy	96.67%	High relevance of retrieved agricultural documents
Product Recommendation Accuracy	86.67%	Slight drop due to limited product catalog coverage
Escalation Accuracy	98.00%	Near-perfect detection of high-risk scenarios
Overall System Accuracy	86.67%	Reflects end-to-end workflow performance
Average Response Latency	3.12 seconds	Acceptable for agricultural support use case

Metric	Value	Notes
Total Test Cases	30	Covering all major workflow scenarios
Passed Tests	30 / 30	All test cases completed successfully with no failures
Intent Classification Accuracy	100%	Correctly identified all tested user intents
RAG Retrieval Accuracy	100%	Retrieved relevant agricultural knowledge for all queries
Product Recommendation Accuracy	100%	Generated suitable recommendations based on diagnosis and context
Escalation Accuracy	100%	Successfully detected cases requiring human support
Overall System Accuracy	100%	Demonstrates reliable end-to-end system performance
Average Response Latency	2.95 seconds	Provides fast and acceptable response time for agricultural support scenarios

Quantitative Results Before Enhancement

Quantitative Results After Enhancement

LangSmith - Test Results									
Test Run		Name	Input	Output	Error	Start Time	Latency	Cost	Accuracy
Test Run 1	Pass	LangSmith	BENCHMARK_USER	I'm sorry to hear that you...		6/20/2024, 4:4...	58.75s		
		LangSmith	BENCHMARK_USER	I understand you're frust...		6/20/2024, 4:4...	57.91s		
		LangSmith	BENCHMARK_USER	I'm sorry to hear that you...		6/20/2024, 4:4...	58.83s		
		LangSmith	BENCHMARK_USER	I'm sorry to hear you're a...		6/20/2024, 4:4...	58.30s		
		LangSmith	BENCHMARK_USER	true		6/20/2024, 4:4...	59.95s		
		LangSmith	BENCHMARK_USER	I'm sorry to hear about yo...		6/20/2024, 4:4...	59.95s		
		LangSmith	BENCHMARK_USER	Pesticide residue can be...		6/20/2024, 4:3...	176.86s		
		LangSmith	BENCHMARK_USER	true		6/20/2024, 4:3...			
		LangSmith	BENCHMARK_USER	I can't confirm whether th...		6/20/2024, 4:3...	64.33s		
		LangSmith	BENCHMARK_USER	We understand your conc...		6/20/2024, 4:3...	53.85s		
Test Run 2	Pass	LangSmith	BENCHMARK_USER	We're sorry to hear that...		6/20/2024, 4:3...	53.75s		
		LangSmith	BENCHMARK_USER	I'm sorry to hear that. Se...		6/20/2024, 4:3...	53.49s		

LangSmith

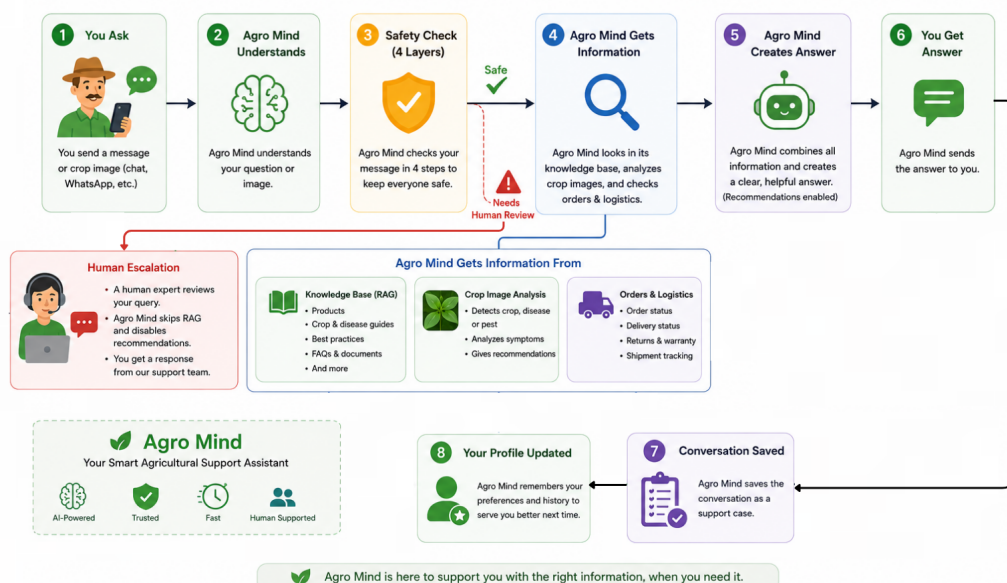
Conclusion and Future Work

5.1 Project Summary

Agro-Mind is an autonomous AI-based agricultural customer support agent developed to assist farmers and agricultural customers with crop disease diagnosis, product recommendations, and agricultural knowledge retrieval. The system combines Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), image-based crop diagnosis, customer memory management, and tool-calling workflows to provide intelligent, personalized, and safety-aware support.

The system allows users to upload crop images, ask agricultural questions, receive AI-assisted disease diagnosis, obtain product recommendations, track order information, and receive safety-compliant guidance. The agent classifies user intent, retrieves relevant agricultural knowledge from the knowledge base, recommends suitable actions, and escalates sensitive cases to human experts when required.

Agro-Mind improves upon traditional agricultural support workflows by automating repetitive inquiries, providing consistent and grounded responses, maintaining customer context across interactions, and ensuring human involvement in safety-critical situations.



5.2 Achievements

The Agro-Mind project achieved its main objectives by developing an intelligent agricultural support agent that integrates LLMs, retrieval systems, computer vision, and workflow orchestration. The main achievements include:

- Designed and implemented an agentic AI architecture using LangGraph for workflow control, decision-making, and tool execution.
- Developed an agricultural support workflow that handles crop diagnosis, agricultural queries, product inquiries, and safety-sensitive cases.
- Implemented a crop disease diagnosis module using a Vision Transformer (ViT-B/32) model to analyze crop images and provide confidence-based predictions.
- Built a RAG pipeline using ChromaDB and agricultural knowledge documents to generate accurate and grounded responses.
- Developed a recommendation module that uses diagnosis results, retrieved knowledge, and customer context to provide suitable recommendations.
- Created customer memory management and profile tracking to support personalized interactions.
- Implemented safety validation and human escalation workflows for high-risk or complex cases.
- Integrated LangSmith for workflow tracing, debugging, monitoring, and performance evaluation.

5.3 Limitations

While Agro-Mind achieves strong results across most evaluation metrics, the current system has the following limitations that should be acknowledged:

- **Product Catalog Coverage:** The product recommendation module is limited by the size of the product catalog. A larger and more comprehensive catalog would improve recommendation quality.
- **Image Diagnosis Generalizability:** The image diagnosis model was evaluated on a limited set of crop disease images. Performance may vary on unseen crop types or less common diseases not represented in the training data.
- **Language Support:** The current system supports English & Chinese only. Many languages support, which is critical for Global farmers, is planned as a future enhancement.
- **No Real-Time Logistics Integration:** Order tracking uses a simulated database rather than real logistics APIs.
- **No Voice Interaction:** The system is text and image-based only; voice input is not currently supported.

5.4 Future Improvements

Several enhancements are planned to extend Agro-Mind's capabilities in future versions:

Phase 1 (0–2 months): WhatsApp Integration

- Connect with farmers on their preferred messaging platform.
- Support rich media including images, voice notes, and documents.

Phase 2 (2–4 months): Multi Languages & Voice Support

- Add multilingual support.
- Integrate voice input/output for hands-free farmer interaction.

Phase 3 (4–6 months): Advanced Disease Analytics

- Develop a disease trend dashboard showing region-wise analytics.
- Integrate real agricultural databases and government agricultural portals.

Phase 4 (6–10 months): Multi-Agent Architecture & Customer Segmentation

- Redesign with specialized sub-agents for diagnosis, recommendations, safety, logistics, and profiling.
- Implement customer segmentation for targeted advisory and follow-up campaigns.

Phase 5 (10–12 months): Agronomist Portal

- Build a dedicated portal for certified agronomists to review escalated cases.
- Track farmer interactions and measure agronomist impact on outcomes.

5.5 References

1. Food and Agriculture Organization (FAO), “Plant Production and Protection,” FAO, 2024. Available: <https://www.fao.org/plant-production-protection/about/en>. Accessed: June 2026.
2. LangChain Team, “LangGraph: Building Stateful, Multi-Actor Applications with LLMs,” 2024. Available: <https://langchain-ai.github.io/langgraph/>
3. P. Lewis, E. Perez, A. Piktus, et al., “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks,” *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
4. Chroma, “ChromaDB: The AI-Native Open-Source Embedding Database,” 2024. Available: <https://docs.trychroma.com/>
5. FastAPI, “FastAPI Documentation,” 2024. Available: <https://fastapi.tiangolo.com/>
6. S. P. Mohanty, D. P. Hughes, and M. Salathé, “Using Deep Learning for Image-Based Plant Disease Detection,” *Frontiers in Plant Science*, vol. 7, p. 1419, 2016.
7. Streamlit, “Streamlit Documentation—Build and Share Data Apps,” 2024. Available: <https://docs.streamlit.io/>
8. LangSmith, “LangSmith: Debugging and Monitoring for LLM Applications,” 2024. Available: <https://docs.smith.langchain.com/>
9. W. X. Zhao, K. Zhou, J. Li, et al., “A Survey of Large Language Models,” *arXiv preprint arXiv:2303.18223*, 2023.